

Joey-Tai Phung

925-348-1457 | jphung@berkeley.edu | [linkedin.com/in/joeyphung/](https://www.linkedin.com/in/joeyphung/) | github.com/joeyphung

WORK EXPERIENCE

Apple Software Engineer | Cupertino, CA

Feb. 2026 - Present

- Developed and maintained automated test infrastructure to validate the functionality, stability, and quality of embedded software across USB, Thunderbolt, PCIe, and Ethernet technologies.
- Served as the primary individual responsible for ensuring PCIe quality, owning testing efforts and collaborating directly with the PCIe host development team to triage bugs and validate PCIe functionality on in-development systems.

SKILLS

Software: Python, C, C++, Go, Java, ROS

Developer Tools: Git, VS Code, Visual Studio, Linux, GDB, Valgrind

EDUCATION

University of California, Berkeley

Aug. 2023 – Dec. 2025

B.S. in Electrical Engineering and Computer Sciences with Highest Honors

GPA: 4.0

PROJECT EXPERIENCE

PintOS | C

Jan. 2025 – May. 2025

- Engineered a system call handler to manage the user-kernel interface, implementing process control (exec, wait, fork), file I/O, and command-line argument passing with pointer validation to ensure kernel integrity.
- Built a preemptive, priority-based thread scheduler with support for priority donation and implemented synchronization primitives including locks, semaphores, and condition variables.
- Developed a thread-safe, hierarchical file system using an indexed inode model for extensible files and a write-back buffer cache to minimize I/O latency.

A Secure File Sharing System | Go

July. 2025

- Architected a secure, multi-user, stateless file-sharing system using layered cryptographic primitives to guarantee data confidentiality and integrity against malicious servers, ensuring seamless and secure synchronization across multiple user devices.
- Devised a secure, segmented file architecture using individually encrypted blocks, a design specifically engineered to enable seamless file sharing, user revocation, and efficient append operations.
- Implemented an access control model with invitation-based sharing and transitive revocation, using key rotation to defend against revoked-user adversaries.

Chinese Checkers Pro | ROS, Python

Oct. 2024 - Dec. 2024

- Developed a Python implementation of Chinese Checkers, incorporating an AI player utilizing Minimax and Alpha-Beta Pruning algorithms, along with an interactive user interface for seamless gameplay and visualization.
- Programmed a Sawyer Robot to accurately detect, classify, pick, and place game pieces using a realsense2 camera, a combination of computer vision techniques and neural networks, AR Tag localization, and MoveIt for arm control.

Build Your Own World | Java

Apr. 2024 - May. 2024

- Implemented algorithms such as Delaunay triangulation and Kruskals for pseudo-random map generation based on user-input seeds.
- Engineered core game mechanics, including saving/loading functionality, user navigation, and interactive features like toggling room and ambient lighting to enhance user engagement.